2024, Mar. 2024 Release LOC:Acute Adult
Acute Coronary Syndrome (ACS)

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Instruction:
While various medical conditions can lead to troponin elevation, these criteria intend to include patients with positive troponins due to acute coronary syndrome (ACS) and those with unstable angina (UA) or suspected ACS. The patient with elevated troponins arising from demand ischemia should be reviewed in the respective subset that corresponds to the source of their demand (e.g., Infection: Sepsis, Infection: Pneumonia, Heart failure, Acute kidney injury, Gastrointestinal (GI) bleeding, Anemia).

Introduction:
Acute coronary syndrome (ACS) refers to a spectrum of conditions that are associated with an abrupt reduction in blood flow to the heart. Conditions include:
- Unstable angina (UA)
- Non-ST-segment elevation myocardial infarction (NSTEMI)
- ST-segment elevation myocardial infarction (STEMI).

At presentation, patients with UA and NSTEMI may be indistinguishable.
UA presents in multiple ways (Knuuti et al., Eur Heart J 2020, 41: 407-77; Owlia et al., J Am Heart Assoc 2019, 8: e012811):
- Rest angina: characteristic pain occurring at rest or for a prolonged period (typically lasting over 20 minutes)
- New onset angina: recent (< 2 months) onset of moderate to severe angina
- Crescendo angina: progressively increases in frequency and intensity, at lower thresholds, over a brief period of time

Acute myocardial infarction (MI), which can present with ST elevations or non-ST elevations on an electrocardiogram (ECG), is defined as a rise and/or fall of troponin above the 99th percentile upper reference limit (URL) together with at least one of the following (Thygesen et al., J Am Coll Cardiol 2018, 72: 2231-64):
- Symptoms of myocardial ischemia
- Development of pathological Q waves
- New ischemic changes on ECG
- Identification of coronary thrombus by angiography
- Imaging evidence of a new loss of myocardium viability or regional wall motion abnormality consistent with

a pattern of ischemia

Universal definitions of myocardial infarction have been classified into the following subtypes (Thygesen et al., J Am Coll Cardiol 2018, 72: 2231-64):

- Type I MI - Caused by coronary artery disease and usually precipitated by acute atherosclerotic plaque disruption causing partial or complete coronary occlusion
- Type II MI - Occurs in the setting of an imbalance between myocardial oxygen demand and supply not related to an acute coronary thrombosis. Management of these patients is primarily directed at treating the underlying etiology (e.g., sepsis, severe anemia, respiratory failure, tachyarrhythmias) (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285; Tamis-Holland et al., Circulation 2019, 139: e891-e908)
- Type III MI - Cardiac death prior to blood sampling or ECG availability but with symptoms suggestive of acute MI or detected by autopsy
- Type IV MI - Directly related to coronary revascularization procedure (e.g., percutaneous coronary intervention (PCI), coronary artery bypass graft surgery (CABG))

Takotsubo syndrome (TTS) - TTS is a cardiomyopathy with a predominance for those assigned the female sex at birth and mortality rates similar to ACS. The presentation is similar to STEMI and exacerbated by extreme physical or emotional stress due to a sharp catecholamine surge. Echocardiography will show signs of left ventricular dysfunction and no coronary artery disease seen on cardiac catheterization (Lyon et al., Eur J Heart Fail 2016, 18: 8-27; Lu et al., ESC Heart Fail 2021, 8: 3663-89)

Evaluation and Treatment:
Management of acute coronary syndrome (ACS) includes rapid evaluation, prompt pharmacological or mechanical reperfusion therapy, and management of possible associated arrhythmias and hemodynamic instability. Patients presenting with pain in the chest, arms, jaw, neck, upper abdomen, shoulder, or other anginal equivalents, such as diaphoresis, shortness of breath, or excessive fatigue, should be evaluated with a 12-lead electrocardiogram (ECG) and troponins. A high index of suspicion of myocardial infarction (MI) is significant in patients with anginal equivalents. A substantial proportion of patients with MI, including older adults, those assigned the female sex at birth, and patients with a history of diabetes and/or heart failure, do not present with classic symptoms of chest pain. Unexplained dyspnea is associated with a high risk of mortality. It is also important to inquire about cocaine or methamphetamine use, which is an additional risk factor for coronary artery disease.
Based on the initial evaluation, an estimation of cardiac risk should be utilized to guide patient placement and management at the appropriate level of care. Treatment consists of standard medical therapy and an individualized management approach. Standard medical therapy is indicated in all patients admitted with a diagnosis of ACS, unless contraindicated, and includes an aspirin, beta blocker, statin, anticoagulant, and antiplatelet therapy.

- Aspirin should be administered as soon as possible and continued indefinitely unless contraindicated. Clopidogrel or prasugrel can be used for those patients who are unable to tolerate aspirin (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285; Ibanez et al., Eur Heart J 2018, 39: 119-77; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).
- Antiplatelet therapy includes the administration of a $P2Y_{12}$ receptor inhibitor (e.g., clopidogrel, prasugrel, ticagrelor) with aspirin or, in some instances, a glycoprotein IIb/IIIa inhibitor infusion (e.g., eptifibatide, tirofiban); (O'Gara et al., Circulation 2013, 127: 529-55). Prasugrel is more effective than clopidogrel but is associated with a significantly higher risk of bleeding (Udell et al., JACC Cardiovasc Interv 2014, 7: 604-12; Orban et al., Thromb Haemost 2014, 112: 1190-7; O'Gara et al., Circulation 2013, 127: e362-425; Melloni et al., 2013 In: Antiplatelet and Anticoagulant Treatments for Unstable Angina/Non-ST Elevation Myocardial Infarction). The administration of both a $P2Y_{12}$ receptor inhibitor and a glycoprotein IIb/IIIa inhibitor is reserved for high-risk patients in whom the benefits outweigh the risk of bleeding, specific rescue situations with the presence of higher intraprocedural thrombus burden or slow-flow closure in stented coronary arteries (Neumann et al., Eur Heart J 2019, 40: 87-165; Ibanez et al., Eur Heart J 2018, 39: 119-77; Amsterdam et al., Circulation 2014, 130: 2354-94).

For patients who are therapeutically anticoagulated upon presentation, initiation of antiplatelet therapy is based on clinical judgment but generally should not be delayed. Patients in whom an initial invasive strategy is selected should receive antiplatelet therapy. Guidelines recommend a proton pump inhibitor (PPI) in combination with dual antiplatelet therapy for patients at increased risk for gastrointestinal bleeding (Ibanez et al., European heart journal 2017, 39: 119-77; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).
In ST-elevation MI (STEMI) cases, improved outcomes rely on delivering reperfusion therapy as rapidly as possible. Reperfusion may be achieved with percutaneous coronary intervention (PCI) (e.g., angioplasty, atherectomy, brachytherapy, with or without stents), thrombolytic therapy, or coronary artery bypass grafting surgery (CABG). The

selection of reperfusion treatment is based on clinical stability, angiographic findings, and local resource availability. PCI is the recommended reperfusion method if experienced providers can perform the procedure promptly. Patients diagnosed with STEMI presenting at a non-PCI-capable hospital should be considered for immediate transfer to a nearby PCI-capable facility following stabilization. PCI should be performed within 12 hours of symptom onset and 90 minutes of first medical contact. Thrombolytic therapy may be appropriate when door-to-balloon time is expected to be greater than 120 minutes and is administered within 12 hours of symptom onset. CABG is not recommended in the acute phase of STEMI but may be appropriate when PCI fails or is not technically feasible due to coronary anatomy, during operative repair of mechanical cardiac defects, or in groups with increased risk for cardiogenic shock or severe heart failure. CABG can be performed within several days of the acute MI in stable patients.
Unstable angina (UA) or non-ST elevation MI (NSTEMI) can be managed with one of two treatment strategies: invasive or selective invasive. The invasive strategy consists of coronary angiography performed within the guideline-recommended 24 hours or earlier of admission based on dynamic ST changes on electrocardiogram, hemodynamic instability, cardiac arrest, acute heart failure from MI, malignant arrhythmias, or refractory symptoms. A selective invasive strategy may be appropriate for patients who are stable and have a low risk of short-term ischemic events; functional tests such as stress echocardiography can be considered for further risk stratification and to guide the need for diagnostic catheterization, which can be performed up to 72 hours of presentation if symptoms recur (Collet et al., Eur Heart J 2021, 42: 1289-367; Ibanez et al., European heart journal 2017, 39: 119-77; van Diepen et al., Circulation 2017, 136: e232-e68; Promes et al., Annals of Emergency Medicine 2017, 70: 724-39; Members et al., Journal of the American College of Cardiology 2017:; Amsterdam et al., Circulation 2014, 130: 2354-94).

InterQual® content contains references to patient sex or gender. Depending on the context, these references may refer to either genotype or phenotype. At the individual patient level, a variety of factors, including but not limited to gender identity and gender affirmation via surgery or hormonal treatment, may affect the applicability of some InterQual criteria. This occurs most often with genetic testing and some procedures that require the presence of specific anatomy. For the purpose of these criteria, references to "Male" and "Female" are based on sex assigned at birth, unless otherwise defined. InterQual users should carefully consider patient genotype and anatomy when appropriate.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the Cochrane Handbook. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources, including the 2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest pain, referred to here as the “2021 American multisociety guideline” (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60; Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285; Winther et al., J Am Coll Cardiol 2020, 76: 2421-32; Knuuti et al., Eur Heart J 2020, 41: 407-77; ; Thygesen et al., Journal of the American College of Cardiology 2018, 72: 2231-64; Ibanez et al., European heart journal 2017, 39: 119-77; Members et al., Journal of the American College of Cardiology 2017:; Roffi et al., Eur Heart J 2016, 37: 267-315; Amsterdam et al., Circulation 2014, 130: 2354-94).

(Excludes PO medications unless noted)

Select Day, One:

- ⊖ **Initial review, One:** (1, 2, 3, 4, 5)
  - (From arrival in ED through decision to admit)
  - (Symptom or finding within 24h)
  - ⊖ **OBSERVATION, One:** (6)
    - ⊖ ACS suspected and, **All:** (7, 8, 9)
      - ⊖ Need for diagnostic cardiac testing based on risk, **One:** (10, 11)
        - − Low risk and ACS related symptoms and age ≥ 40 years (7, 8, 12, 13, 14)
        - ⊖ Low risk and dyspnea and, **One:** (12, 13, 14, 15)
          - − Male and age ≥ 40 years (16)
          - − Female and age ≥ 50 years (17)
        - − Intermediate risk (11, 18, 19)
      - − ECG normal, unchanged, or non-diagnostic (20, 21)
      - − Initial troponin negative or indeterminate and requiring serial troponins (22)
      - ⊖ Intervention scheduled or performed within 24h, **One:** (23)
        - − Cardiac catheterization (24)
        - − CCTA (25)
        - − Stress test (26)
  - ⊖ **INTERMEDIATE, One:** (27)
    - ⊖ Acute chest pain or anginal equivalent and, **Both:** (7, 8)
      - ⊖ Finding, ≥ **One:** (28)
        - − High risk by Clinical Decision Pathway (11, 29, 30)
        - − New diagnosis of moderate-severe ischemia on stress test (31)
        - − New ischemic changes on ECG (32, 33)
        - − New onset left ventricular dysfunction by echocardiogram with ejection fraction ≤ 40%(0.40)
        - − Troponin > ULN and > baseline (34)
      - ⊖ Intervention, ≥ **One:**
        - − Anticoagulation (35)
        - − Cardiac catheterization scheduled within 24h (24)
    - ⊖ NSTEMI and pain resolved or controlled with medication and, ≥ **One:** (36, 37, 38)
      - − Anticoagulation (35)
      - − Cardiac catheterization scheduled within 24h (24)
    - ⊖ Unstable angina controlled with medication and, **All:** (8, 37, 38, 39, 40, 41)
      - − ECG with new or worsening ischemia or pacemaker rhythm (42, 43, 44)
      - − Initial troponin negative or indeterminate and requiring serial troponins (22)
      - ⊖ Intervention, ≥ **One:** (35, 45)
        - − Anticoagulation (35)
        - − Cardiac catheterization scheduled within 24h (24)
  - ⊖ **CRITICAL, ≥ One:** (46)
    - − ACS suspected and new LBBB on ECG (47)
    - ⊖ Arrhythmia, ischemic and, ≥ **One:** (48)
      - − Urgent electrical cardioversion
      - − Defibrillation performed within last 24h
      - − Temporary pacemaker insertion performed within last 2d
      - − Antiarrhythmic
    - ⊖ NSTEMI and persistent chest pain or anginal equivalent and, **One:** (7, 8, 36, 38, 41)
      - ⊖ Anticoagulation and, ≥ **One:** (35, 45)
        - − Cardiac catheterization scheduled within 24h (24)
        - − Nitroglycerin, continuous (49)
    - − STEMI (50)
    - ⊖ Unstable angina and persistent chest pain or anginal equivalent and, **All:** (7, 8, 38, 39, 40, 41)
      - − ECG with new or worsening ischemia or pacemaker rhythm (42, 43, 44)

- Initial troponin negative or indeterminate and requiring serial troponins [22]
- ⊖ Anticoagulation and, ≥ **One:** [35, 45]
  - Cardiac catheterization scheduled within 24h [24]
  - Nitroglycerin, continuous [49]

- ⊖ **Episode Day 1, One:** [1, 2, 4, 5]
  (Symptom or finding within 24h)
  - ⊖ **OBSERVATION**, **One:** [6]
    - ⊖ ACS suspected and, **All:** [7, 8, 9]
      - ⊖ Need for diagnostic cardiac testing based on risk, **One:** [10, 11]
        - Low risk and ACS related symptoms and age ≥ 40 years [7, 8, 12, 13, 14]
        - ⊖ Low risk and dyspnea and, **One:** [12, 13, 14, 15]
          - Male and age ≥ 40 years [16]
          - Female and age ≥ 50 years [17]
        - Intermediate risk [11, 18, 19]
      - ECG normal, unchanged, or non-diagnostic [20, 21]
      - Initial troponin negative or indeterminate and requiring serial troponins [22]
      - ⊖ Intervention scheduled or performed within 24h, ≥ **One:** [23]
        - Cardiac catheterization [24]
        - CCTA [25]
        - Stress test [26]
  - ⊖ **INTERMEDIATE**, **One:** [27]
    - ⊖ Acute chest pain or anginal equivalent and, **Both:** [7, 8]
      - ⊖ Finding, ≥ **One:**
        - High risk by Clinical Decision Pathway [11, 28, 29]
        - New diagnosis of moderate-severe ischemia on stress test [31]
        - New ischemic changes on ECG [32, 33]
        - New onset left ventricular dysfunction on echocardiogram with ejection fraction ≤ 40%(0.40)
        - Troponin > ULN and > baseline [34]
      - ⊖ Intervention, ≥ **One:**
        - Anticoagulation [35]
        - Cardiac catheterization scheduled or performed within 24h [24]
    - ⊖ NSTEMI and pain resolved or controlled with medication and, ≥ **One:** [36, 37, 38]
      - Anticoagulation [35]
      - Cardiac catheterization scheduled or performed within 24h [24]
    - ⊖ Unstable angina controlled with medication and, **All:** [8, 37, 38, 39, 40, 41]
      - ECG with new or worsening ischemia or pacemaker rhythm [42, 43, 44]
      - Initial troponin negative or indeterminate and requiring serial troponins [22]
      - ⊖ Intervention, ≥ **One:** [35, 45]
        - Anticoagulation [35]
        - Cardiac catheterization scheduled or performed within 24h [24]
  - ⊖ **CRITICAL, ≥ One:** [46]
    - ACS suspected and new LBBB on ECG [47]
    - ⊖ Arrhythmia, ischemic and, ≥ **One:** [48]
      - Urgent electrical cardioversion
      - Defibrillation performed within last 24h
      - Temporary pacemaker insertion performed within last 2d
      - Antiarrhythmic
    - ⊖ NSTEMI and persistent chest pain or anginal equivalent and, **One:** [7, 8, 36, 38, 41]
      - ⊖ Anticoagulation and, ≥ **One:** [35, 45]
        - Cardiac catheterization performed within 24h [24]
        - Nitroglycerin, continuous [49]
    - STEMI [50, 51]
    - ⊖ Unstable angina and persistent chest pain or anginal equivalent and, **All:** [7, 8, 38, 39, 40, 41]

- ECG with new or worsening ischemia or pacemaker rhythm [42, 43, 44]
- Initial troponin negative or indeterminate and requiring serial troponins [22]
- ⊖ Anticoagulation and, ≥ **One:** [35, 45]
  - Cardiac catheterization performed within 24h [24]
  - Nitroglycerin, continuous [49]
- ECMO or ECLS [52]
- Hemodynamic monitoring, invasive [53]
- IABP [54, 55]
- Mechanical ventilation
- Thrombolytic therapy [56]
- Vasopressor, vasodilator (excludes nitroglycerin), or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, percutaneous [57, 58]

⊖ **Episode Day 2, One:** [4, 5, 59]
- ⊖ **OBSERVATION, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **One:** [60]
    - ⊖ ACS ruled out and, **Both:**
      - Troponins negative for cardiac injury
      - ⊖ Objective cardiac risk assessment, **One:** [61, 62]
        - Completed prior to discharge [63]
        - Low risk and scheduled outpatient cardiac testing [14, 64]
  - ⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
    - For a condition other than ACS use appropriate subset (Episode Day 1) based on clinical findings
    - For ACS and abnormal diagnostic testing, use Episode Day 2 at a higher level of care [65]
    - Refer for secondary review
- ⊖ **INTERMEDIATE, One:**
  - ⊖ Abnormal cardiac diagnostic testing post-Observation care and, ≥ **One:** [25, 65, 66, 67]
    - Anticoagulation [35]
    - Cardiac catheterization scheduled or performed within 24h [24]
  - ⊖ High risk and, ≥ **One:** [68]
    - Anticoagulation
    - Cardiac catheterization scheduled or performed within 24h [24]
  - ⊖ NSTEMI and pain resolved or controlled with medication and, ≥ **One:** [36, 37, 38]
    - Anticoagulation [35]
    - Cardiac catheterization scheduled or performed within 24h [24]
  - STEMI [51, 69]
  - ⊖ Unstable angina controlled with medication and, ≥ **One:** [8, 37, 38, 39, 40]
    - Anticoagulation [35]
    - Cardiac catheterization scheduled or performed within 24h [24]
  - Post cardiac catheterization ≤ 24h [70, 71]
  - Post Critical care ≤ 24h
- ⊖ **CRITICAL, ≥ One:**
  - Acute kidney injury (use **Acute kidney injury** subset)
  - Non-ischemic atrial arrhythmia (use **Arrhythmia: Atrial** subset)
  - Non-ischemic heart block (use **Arrhythmia: Blocks** subset)
  - Non-ischemic ventricular arrhythmia (use **Arrhythmia: Ventricular or abnormal ECG finding** subset)
  - Pericardial effusion (use **General Medical** subset)
  - ⊖ NSTEMI and persistent chest pain or anginal equivalent and, **One:** [7, 8, 36, 38, 41]
    - ⊖ Anticoagulation and, ≥ **One:** [35, 45]
      - Cardiac catheterization performed within 24h [24]
      - Nitroglycerin, continuous [49]
  - ⊖ Unstable angina and persistent chest pain or anginal equivalent and, **All:** [7, 8, 38, 39, 40, 41]

- ECG with new or worsening ischemia or pacemaker rhythm [42, 43, 44]
- Initial troponin negative or indeterminate and requiring serial troponins [22]
- ⊖ Anticoagulation and, ≥ **One:** [35, 45]
  - Cardiac catheterization performed within 24h [24]
  - Nitroglycerin, continuous [49]

- ⊖ Arrhythmia, ischemic and, ≥ **One:** [48]
  - Urgent electrical cardioversion performed within last 24h
  - Defibrillation performed within last 24h
  - Temporary pacemaker insertion performed within last 2d
  - Antiarrhythmic
- ECMO or ECLS [52]
- Heart failure and diuretic every 1-2h or continuous
- Hemodynamic monitoring, invasive [53]
- IABP [54, 55]
- Mechanical ventilation
- ⊖ NIPPV and, ≥ **One:** [72]
  - FiO_2 > 50%(0.50)
  - Respiratory intervention every 1-2h [73]
- Vasopressor, vasodilator (excludes nitroglycerin), or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- VAD, percutaneous [57, 58]

⊖ **Episode Day 3, One:** [4, 5]

- ⊖ **ACUTE, One:** [74]
  - Acute heart failure (use **Heart Failure** subset)
  - Acute kidney injury (use **Acute kidney injury** subset)
  - COPD (use **COPD** subset)
  - Bleeding (use **Non-traumatic Bleeding** subset)
  - Gastrointestinal bleeding (use **Gastrointestinal Bleeding** subset)
  - STEMI [51, 69]
- ⊖ **INTERMEDIATE, One:**
  - ⊖ **Responder,** discharge expected today if clinically stable last 12h, **All:** [60]
    - Blood pressure within acceptable limits [75, 76]
    - Heart rate within acceptable limits [75]
    - No evidence of worsening ischemia on ECG within last 24h
    - Pain resolved
    - ⊖ Complication or comorbidity, **One:** [77]
      - No complication or active comorbidity relevant to this episode of care [77]
      - ⊖ Complication or comorbidity and clinically stable for discharge, ≥ **One:**
        - Arrhythmia controlled or resolved
        - Anticoagulation regimen established [78, 79]
  - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - Pericardial effusion (use **General Medical** subset)
    - ⊖ NSTEMI and chest pain resolved or controlled with medication and, ≥ **One:** [36, 37, 38]
      - Anticoagulation [35]
      - Cardiac catheterization performed within 24h [24]
    - ⊖ Abnormal vital signs ≤ 48h, **Both:**
      - ⊖ Finding, ≥ **One:**
        - Heart rate < 60/min or > 100/min
        - Systolic blood pressure > 140 mmHg
        - Diastolic blood pressure > 100 mmHg
      - Medication adjustment (includes PO) [80]
    - ⊖ Arrhythmia, ischemic and, ≥ **One:**
      - Antiarrhythmic and hemodynamically stable

- Conversion from IV to PO antiarrhythmic and no proarrhythmic risk ≤ 24h
- Initiation of PO antiarrhythmic with proarrhythmic risk ≤ 3d [81, 82, 83]
- Intracardiac thrombus and anticoagulation (includes PO) [84]
- NIPPV [72]
- Oxygen ≥ 40%(0.40), ≤ 2d [85]
- Permanent pacemaker or ICD placement performed within last 24h [86]
- Post cardiac catheterization ≤ 24h [70, 71]
- Post Critical care ≤ 24h

- **CRITICAL,** ≥ **One:**
  - Pericardial effusion (use **General Medical** subset)
  - Arrhythmia, ischemic and, ≥ **One:** [48]
    - Urgent electrical cardioversion performed within last 24h
    - Defibrillation performed within last 24h
    - Temporary pacemaker insertion performed within last 2d
    - Antiarrhythmic
  - ECMO or ECLS [52]
  - Heart failure and diuretic every 1-2h or continuous
  - Hemodynamic monitoring, invasive [53]
  - IABP [54, 55]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [72]
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h [73]
  - Vasopressor, vasodilator (excludes nitroglycerin), or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
  - VAD, percutaneous [57, 58]
- **Episode Day 4, One:** [4, 5]
  - **ACUTE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [60]
      - Blood pressure within acceptable limits [75, 76]
      - Heart rate within acceptable limits [75]
      - No complication or active comorbidity relevant to this episode of care [77]
      - No evidence of worsening ischemia on ECG within last 24h
      - Objective cardiac risk assessment complete [62]
      - Pain resolved
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - **INTERMEDIATE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [60]
      - Blood pressure within acceptable limits [75, 76]
      - Complication or comorbidity, **One:** [77]
        - No complication or active comorbidity relevant to this episode of care [77]
        - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - Arrhythmia controlled or resolved
          - Anticoagulation regimen established [78, 79]
      - Heart rate within acceptable limits [75]
      - No evidence of worsening ischemia on ECG within last 24h
      - Pain resolved
    - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
      - Abnormal vital signs ≤ 48h, **Both:**
        - Finding, ≥ **One:**

- Heart rate < 60/min or > 100/min
- Systolic blood pressure > 140 mmHg
- Diastolic blood pressure > 100 mmHg
- Medication adjustment (includes PO) [80]
- Arrhythmia, ischemic and, ≥ **One:**
  - Antiarrhythmic and hemodynamically stable
  - Conversion from IV to PO antiarrhythmic and no proarrhythmic risk ≤ 24h
  - Initiation of PO antiarrhythmic with proarrhythmic risk ≤ 3d [81, 82, 83]
- Intracardiac thrombus and anticoagulation (includes PO) [84]
- NIPPV [72]
- Oxygen ≥ 40%(0.40), ≤ 2d [85]
- Permanent pacemaker or ICD placement performed within last 24h [86]
- Post Critical care ≤ 24h

- **CRITICAL, ≥ One:**
  - Arrhythmia, ischemic and, ≥ **One:** [48]
    - Urgent electrical cardioversion performed within last 24h
    - Defibrillation performed within last 24h
    - Temporary pacemaker insertion performed within last 2d
    - Antiarrhythmic
  - ECMO or ECLS [52]
  - Heart failure and diuretic every 1-2h or continuous
  - Hemodynamic monitoring, invasive [53]
  - IABP [54, 55]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [72]
    - FiO_2 > 50%(0.50)
    - Respiratory intervention every 1-2h [73]
  - Vasopressor, vasodilator (excludes nitroglycerin), or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
  - VAD, percutaneous [57, 58]
- **Episode Day 5, One:**
  - **INTERMEDIATE, One:**
    - **Responder,** discharge expected today if clinically stable last 12h, **All:** [60]
      - Blood pressure within acceptable limits [75, 76]
      - Complication or comorbidity, **One:** [77]
        - No complication or active comorbidity relevant to this episode of care [77]
        - Complication or comorbidity and clinically stable for discharge, ≥ **One:**
          - Arrhythmia controlled or resolved
          - Anticoagulation regimen established [78, 79]
      - Heart rate within acceptable limits [75]
      - No evidence of worsening ischemia on ECG within last 24h
      - Pain resolved
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - **CRITICAL, One:**
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review

Discharge, Level of Care, One: [87]

- ⊖ **HOME, All:**
  - ⊖ Level of care appropriateness, **Both:**
    - Home environment safe and accessible [88]
    - Patient or caregiver demonstrates ability to manage care
  - ⊖ ACS discharge planning, **Both:**
    - Provide comprehensive written discharge and teaching instructions [89]
    - ⊖ Education on the importance of, **All:**
      - Medication adherence [90]
      - Recognizing acute cardiac symptoms and appropriate actions [91]
      - Secondary prevention methods [92, 93]
      - Physical activity [94]
      - Recognizing signs of depression [95]
  - Medication reconciliation [96]
  - Follow-up care planned [97]
  - Identify and address transportation needs
- ⊖ **HOME CARE, All:**
  - ⊖ Level of care appropriateness, **All:**
    - Home environment safe and accessible [88]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [98, 99]
    - ⊖ Skilled services, ≥ **One:**
      - Skilled nursing [100]
      - Skilled therapy, PT, OT, or ST
      - Behavioral health [101]
  - ⊖ ACS discharge planning, **Both:**
    - Provide comprehensive written discharge and teaching instructions [89]
    - ⊖ Education on the importance of, **All:**
      - Medication adherence [90]
      - Recognizing acute cardiac symptoms and appropriate actions [91]
      - Secondary prevention methods [92]
      - Physical activity [94]
      - Recognizing signs of depression [95]
  - Medication reconciliation [96]
  - Follow-up care planned [97]
  - Identify and address transportation needs
- ⊖ **BEHAVIORAL HEALTH, Both:**
  - ⊖ Level of care appropriateness, **One:**
    - Support systems available [102]
  - ⊖ Skilled treatment, ≥ **One:**
    - Clinical assessment [103]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [104]
- ⊖ **SKILLED MEDICAL OR THERAPY, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [105]
      - ⊖ Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [106, 107]
        - Functional impairments requiring at least supervision [108]
        - Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
  - ⊖ Complete prior to facility transfer, **All:**

- Medication reconciliation [96]
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

- **SUBACUTE MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [105]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [106, 107]
        - ≥ 2 functional impairments requiring at least minimum assistance [108]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [96]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **SUBACUTE REHAB, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [106, 107]
      - ≥ 2 functional impairments requiring at least minimum assistance [108]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [96]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **ACUTE REHAB, Both:**
  - Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Services, **All:**
      - Goal directed therapy with rehabilitation potential [106, 107]
      - ≥ 2 functional impairments requiring at least minimum assistance [108]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [109]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [96]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- **LONG TERM ACUTE CARE, Both:**
  - Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:** [110]
      - Primary condition or illness and treatment of active comorbid condition(s) [111]
      - Ventilator dependent ≥ 6h/d and weaning planned [112]
  - Complete prior to facility transfer, **All:**
    - Medication reconciliation [96]
    - Obtain and complete forms for facility

- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
280 ACUTE MYOCARDIAL INFARCTION, DISCHARGED ALIVE WITH MCC	**n/a**	**4.2**	**CMS GMLOS**
281 ACUTE MYOCARDIAL INFARCTION, DISCHARGED ALIVE WITH CC	**n/a**	**2.4**	**CMS GMLOS**
282 ACUTE MYOCARDIAL INFARCTION, DISCHARGED ALIVE WITHOUT CC/MCC	**n/a**	**1.8**	**CMS GMLOS**
311 ANGINA PECTORIS	**n/a**	**2**	**CMS GMLOS**
NSTEMI	**8%**	**3.6**	**InterQual**
STEMI	**2%**	**3**	**InterQual**
UNSTABLE ANGINA	**67%**	**1.8**	**InterQual**

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:

1:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Transition Plan: InterQual® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. Patients with acute coronary syndrome (ACS) may be at high risk for readmission and careful attention to discharge planning is essential. Despite 30-day readmissions on a downward trend from 2007 to 2017, readmission rates after Myocardial infarction (MI) still trend at 11-16% even with the hospital readmissions reduction program (HRRP) in place. Readmission rates are higher for patients with other co-morbidities, including age over 75, chronic obstructive pulmonary disease (COPD), chronic kidney disease, heart failure, stroke, peripheral vascular disease, and diabetes (Tsao et al., Circulation 2022, 145: e153-e639; Tisminetzky et al., Am J Med 2021, 134: 1127-34; Ko et al., J Am Coll Cardiol 2020, 75: 736-46; Wang et al., Clin Cardiol 2019, 42: 889-98; Andell et al., Open Heart 2014, 1: e000002).

3:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

4:

Guideline Standard Considerations

- Beta blocker
- Aspirin
- Antiplatelet
- Anticoagulant
- Angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB) if heart failure present
- Statin
- Supplemental oxygen for O2 saturation below 90%(0.90)
- Nitrates for ongoing chest pain (Amsterdam et al., Circulation 2014, 130: 2354-94)

Expected progress after 24h

- Angina improving
- Electrocardiogram (ECG) improving
- Troponins trending down
- Vital signs stable or within acceptable limits

Consider:

- Decreasing or discontinuing oxygen
- Increasing activity
- Continuous cardiac monitoring
- Transition to PO medications
- Cardiac rehabilitation evaluation
- Nutrition consult for cardiac diet education
- Smoking cessation education if applicable
- Discharge planning

5:

Care Facilitation

Referral for psychiatric evaluation and/or treatment if co-occurring psychiatric symptoms are suspected or confirmed (e.g., depression, anxiety) **(Home Care)**

- Knowledge deficit or non-adherence with condition management identified
- Functional and/or cognitive impairment requiring additional therapy

Potential risk of readmission **(Home Care, Readmission reduction team)**

- Knowledge deficit or non-adherence with condition management identified
- Co-morbid hypertension, heart failure, diabetes, chronic kidney disease, atrial fibrillation, obesity, smoking
- Myocardial infarction admission within the last year

Functional or activity of daily living (ADL) impairment - consider an alternate level of care for therapy needs

- Physical therapy provided in the home setting should be considered if the home is safe and accessible **(Home Care)**

- If home environment is unsafe, consider skilled nursing facility **(SNF)**

End of life planning
- Consider palliative care for patients for end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Social Determinants of Health (SDOH)
Social concerns may impact the decision to admit, care coordination during hospital stay, as well as discharge destination. The following social determinants of health should be evaluated on admission to determine which factors are relevant to an individual patient and condition. This list may not be all inclusive and all relevant SDOH factors should be considered. Any SDOH factors determined to be relevant to a given patient should be planned for throughout the hospital stay.
Alcohol or substance use
- Care coordination with outpatient medical or behavioral health providers
- Information should be given about community resources and programs
- Referral to addiction social support systems

Domestic violence and sexually abused individuals
- Community resources and legal information should be given
- Create a long-term safety plan or make appropriate referral

End of life planning, arrange palliative care or hospice
- Consider palliative care for patients with end stage disease
- Consider hospice care for terminally ill patients with a life expectancy of less than 6 months

Financial resources strained or limited
- For patients with income insecurity, identify income-support resources (Pottie et al., CMAJ 2020, 192: E240-E54)
- Consider direct income assistance benefits and indirect income assistance programs (i.e., programs that improve access to basic living necessities such as provision of food, daycare, and fuel or rent supplements)
- Consider referral to Area Agencies on Aging (AAA) for information on nutrition and meals programs, home homemaking services, transportation resources
- For food insecurity, provide information about local food banks and Supplemental Nutrition Assistance Program (SNAP) incentive programs (Berman et al., Pediatr Rev 2018, 39: 235-46)

Functional or ADL impairment
- Consider an alternate level of care
- If patient is unable to make decision for self, refuses treatment, or demands to be discharged without a rational reason, a capacity evaluation may be needed to ensure patient's ability to make decisions for self
- Information should be given to patient and family about Advance Directive or Power of Attorney

Mental health co-morbidities involved
- Appointment with primary care provider or behavioral health specialist upon discharge
- Care coordination with outpatient behavioral health providers
- Community resources should be given

Risk for readmission
- Consider home care services
- Consider transfer to an alternative level of care or readmission reduction program
- Provide patient and family with information on Disease Management programs
- Consider telehealth and mobile health interventions

Risk for treatment nonadherence
- For patients with poor insight into illness or noncompliance with treatment, introduce peer support groups to help improve service user experience and quality of life (National Institute for Health and Care Excellence (NICE). Psychosis and schizophrenia in adults: prevention and management. CG178. London: NICE; 2014.)
- In-home intervention to prevent medical relapse, or readmission, and to enhance medication compliance

- For patients at risk of medication nonadherence, consider patient education and behavioral support initiatives such as text message reminders to take a medication and electronic medication dispensers.

Environmental exposure to toxic substances and other physical hazards

- For tobacco use and/or exposure information should be given about tobacco-use cessation and the risk of second-hand smoke
- For lead poisoning, provide community resources for primary prevention and secondary prevention strategies, i.e., blood lead level testing and follow up (Centers for Disease Control and Prevention, Lead Poisoning Prevention. 2019)

Unstable living environment or homelessness expected upon discharge

- Health insurance application
- Homeless shelter application and information should be given to patient
- Referral to walk-in state or federal sponsored outpatient programs
- Ensure access to local housing coordinator or case manager for immediate link to permanent supportive housing and coordinated access system (Pottie et al., CMAJ 2020, 192: E240-E54)
- For individuals with homelessness and the greatest service need (i.e., serious mental illness, history of hospitalizations, functional impairment) consider Intensive Case Management Intervention, Assertive Community Treatment, or other Intensive Community-Based Treatment (Ponka et al., PLoS One 2020, 15: e0230896)

Unsafe home environment

- If home environment is unsafe, consider a skilled nursing facility (SNF)
- For patients at risk for falls, provide information on community-based fall prevention programs through the National Council on Aging

6:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

7:

This criterion refers to acute chest pain or discomfort, which can be described by either new onset or a change in pattern, intensity, or duration from previous episodes in a patient with recurrent symptoms.

Suspected cardiac chest pain can be described as (but not limited to) pain, pressure, tightness, squeezing, discomfort, heaviness, and burning; it is more typically exertional and retrosternal in location, although patients may report a location of pain other than the chest including the shoulder, jaw, upper abdomen, and back. The 2021 American guidelines discourage the use of “typical” and “atypical” to characterize chest pain and instead advocate for describing chest pain as either cardiac, possibly cardiac, or noncardiac (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

8:

This criterion refers to patients with symptoms such as angina or anginal equivalents suspected to be secondary to acute coronary syndrome (ACS). Approximately one-third of patients experiencing ACS may not have classic left-sided chest pain or discomfort symptoms. It is essential to consider other signs or symptoms representing cardiac causes of suspected ACS. We refer to this as anginal equivalent, especially when evaluating a patient with risk factors or a history of coronary artery disease. These presentations are seen more frequently in females assigned at birth, older adults, and people with diabetes. Symptoms may include nausea with vomiting, new onset or increased dyspnea, unexplained fatigue, diaphoresis, and syncope (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285; Ibanez et al., Eur Heart J 2018, 39: 119-77; Amsterdam et al., Circulation 2014, 130: e344-426; O'Gara et al., Circulation 2013, 127: 529-55)

9:

A suspicion of acute coronary syndrome (ACS) is based on evidence of angina or anginal equivalent suggesting acute myocardial ischemia. Depending on the clinical suspicion by history, electrocardiogram (ECG) findings, risk factors, and troponins, patients can be assessed using guideline-recommended clinical decision pathways (CDP) to determine the short-term risk of adverse cardiac events. Patients with suspected ACS deemed low or intermediate risk are candidates for Observation level of care. During this time, patients are evaluated, symptoms are monitored and observed, and serial troponin testing and ECGs are assessed to determine if acute myocardial ischemia is present. This, in turn, will help the clinician determine the need for any further cardiac testing (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

10:

This criterion refers to the risk of having index event myocardial infarction (MI) or a major adverse cardiovascular event (MACE) within 30 days, outlined in both the 2021 American multisociety and European Society of Cardiology (ESC) guidelines. MACE, commonly used as an endpoint to measure specific clinical cardiac outcomes, can include all-cause and cardiac-related mortality, myocardial infarction, and/or need for revascularization(Collet et al., Eur Heart J 2021, 42: 1289-367; Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285). The 30-day MACE risk can be determined by either high-sensitivity troponins alone or clinical decision pathways (CDPs), primarily if conventional troponins are utilized (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285). Per the 2021 American multisociety guidelines, the most widely studied and validated of these risk scores include:

- HEART score
- TIMI score (as used in the ADAPT trial)

CDPs provide structured validated protocols to evaluate patients suspected of having acute coronary syndrome (ACS) using troponin measurements at specified time intervals and evaluation of electrocardiogram (ECG) to categorize patients as low, intermediate, or high risk. Guidelines recommend routine and standardized use of CDPs in the emergency departments and ambulatory settings to assist in subsequent management (e.g., testing) and safe disposition, especially of low-risk patients when clinically appropriate (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60; Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285) CDPs all perform well to define the risk of 30-day MACE and index visit acute myocardial infarction (MI). The efficacy of CDPs improves with the use of high-sensitivity troponins (hs-cTn) compared to conventional troponins; this combination has been shown to improve earlier detection of AMI, reduce testing, and increase emergency department (ED) discharges when appropriate. CDPs have also been shown to reduce unnecessary testing and hospital admissions while maintaining higher sensitivity for 30-day MACE. The American College of Emergency Physicians (ACEP) recommends the use of HEART or TIMI scores integrated with hs-cTn to predict rates of 30-day MACE (Sandoval et al., Circulation 2022, 146: 569-81; Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60; Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

11:

The HEART (History, ECG, Age, Risk factors, Troponin) score was first developed in 2008 as a diagnostic aid for rapid risk stratification of patients presenting to the emergency departments (ED) with symptoms related to acute coronary syndrome (ACS) from readily available clinical data. Over the years, this tool has proved to be an accurate and valid diagnostic tool for the probability or risk of major adverse cardiovascular events (MACE) within 4-6 weeks after index presentation. The HEART score effectively predicted all-cause mortality, myocardial infarction (MI), or the need for coronary revascularization (Januzzi et al., J Am Coll Cardiol 2019, 73: 1059-77). In a randomized controlled trial conducted in 2015, the use of the HEART pathway (HEART score and troponin at 0h and 3h intervals) led to a statistically significant reduction in median length of stay by 12 hours, objective cardiac testing by 12%, and hospitalizations by 21% (Mahler et al., Circ Cardiovasc Qual Outcomes 2015, 8: 195-203). The study showed that when the HEART score of 3 or less (low-risk group) was combined with negative troponins, the results showed a negative predictive value (NPV) exceeding 99% and sensitivity of 98.3% for 30-day MACE when compared with the use of either negative hs-cTn or nonischemic electrocardiogram (ECG) alone. For the group of low-risk patients, the HEART pathway allowed for early rule out of MI and safe discharge without a single occurrence of 30-day MACE. Follow-up studies published by the same lead author in 2018 showed that among those patients identified as low risk by the HEART pathway, 0.4% of 1461 patients experienced MACE within 30 days (2 of these were MI with a miss rate of 0.1%).
In the 2013 prospective study conducted by Backus et al., patients were assigned HEART scores based on the findings at ED presentation. The primary endpoints were MACE within six weeks. The results showed the following MACE outcomes as listed below (Backus et al., Int J Cardiol 2013, 168: 2153-8):

- HEART score ≤ 3 (low): 1.7%

- HEART score 4-6 (intermediate): 16.6%
- HEART score ≥ 7 (high): 50.1%

HEART score is determined by tabulating the applicable points (0-2) from the five components (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285; Mahler et al., Circulation 2018, 138: 2456-68; Mahler et al., Circ Cardiovasc Qual Outcomes 2015, 8: 195-203):

- History:
    - 0: slightly suspicious
    - 1: moderately suspicious
    - 2: highly suspicious

- ECG:
    - 0: normal
    - 1: nonspecific changes
    - 2: changes consistent with ACS

- Age:
    - 0: < 45 years
    - 1: 45-65 years
    - 2: > 65 years

- Risk factors (diabetes, tobacco use, hypertension, hyperlipidemia, obesity, family history of coronary artery disease (CAD):
    - 0: no risk factors
    - 1: 1-2 risk factors
    - 2: ≥3 risk factors

- Troponin (initial):
    - 0: normal
    - 1: > 1 to < 3x upper limit of normal (ULN)
    - 2: ≥ 3x ULN

12:

This criterion refers to those patients classified as low-risk who may need to undergo further cardiac testing. Low-risk for patients presenting with acute chest pain is defined by any one of the following criteria based on troponins or clinical decision pathways (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285):

- HEART score ≤ 3 with negative serial troponins (high-sensitivity troponins (hs-cTn) at 0 and 1-3 hours or conventional troponins (cTn) at 0 and 3-6 hours)
- TIMI 0-1 (ADAPT study) with negative hs-cTn or cTn at 0h and 2 hours
- Negative (very low threshold) hs-cTn at 0 hours (on presentation) if symptom onset > 3 hours
- Negative (low threshold) hs-cTn at 0 and 2 hours

The 2021 American multisociety guidelines for the evaluation of chest pain recommend the use of the clinical decision pathways (CDPs) to assess the level of risk (low, intermediate, high) to determine the risk of index myocardial infarction (MI) and 30-day major adverse cardiovascular events (MACE). Patients are deemed low-risk if their probability of 30-day MACE is less than 1% (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285). A recently published US multisite study compared a single hs-cTn on presentation to serial hs-cTn protocol at 0 hours and 1 to 3 hours for patients presenting to the emergency department (ED) with suspected acute coronary syndrome (ACS) to determine the negative predictive value of 30-day MACE or index visit MI (Allen et al., Circulation 2021, 143: 1659-72). This study of 1463 patients evaluated the primary outcome of MACE in 30-days including MI during the index visit, and found that patients tested with only an initial hs-cTn vs. serial 0-hour and 1-hour algorithm were found to have a negative predictive value (NPV) of 98.4% and 97.2%, respectively. The writers then added nonischemic electrocardiograms (ECG) into each group, resulting in no change to the NPV or sensitivities. Finally, HEART scores of 3 or below, classified as low risk, were factored into both arms, and results showed a slight but measurable increase in NPV and sensitivity for each group. When low-risk HEART scores were added, NPVs increased to 99% for the initial hs-cTn group versus 98.4% for the group utilizing serial 0-hour and 1-hour hs-cTn protocol. According to Society for Academic Emergency Medicine (SAEM) and the American College of Emergency Physicians (ACEP), an NPV of less than 99% was deemed unacceptable for a 30-day MACE.

Based on the 2021 American multisociety guidelines, there are no clear recommendations for patients with acute chest pain who meet low-risk scores by CDP but may have baseline cardiac risk factors that could warrant further evaluation for possible underlying obstructive coronary artery disease. The previous 2014 Americal College of Cardiology guidelines recommended stress testing within 72 hours after the index ED visit for chest pain after ACS was ruled out (Amsterdam et al., Circulation 2014, 130: e344-426). However, the most recent guidelines suggest that low-risk score patients with chest pain can be discharged without urgent cardiac testing and with outpatient follow-up. Additional guidance for outpatient cardiac testing covers patients with chronic or stable chest pain but not patients who may have had acute chest pain, presented to the ED, and were found to be at low risk of 30-day MACE. InterQual® expert peer reviewers agree that some patients with acute chest pain deemed low-risk for 30-day MACE may warrant further noninvasive testing based on their Pre-Test Probability (PTP) of having obstructive coronary artery disease (CAD). We recognize that some patients may have barriers to obtaining timely noninvasive cardiac testing within 30 days from challenges surrounding access to healthcare. Hence, it may be necessary to further stratify the low-risk scoring patients with a nonischemic ECG and negative hs-cTn by incorporating PTP scores to allow Observation level of care for noninvasive testing. This population of patients would be categorized as having a pretest probability of CAD of 6% or higher. Most patients with a very low PTP of CAD (≤ 5%) based on European Society of Cardiology (ESC) guidelines did not warrant further cardiac testing and were deemed safe for discharge from the ED (Knuuti et al., Eur Heart J 2020, 41: 407-77). Also, based on these guidelines, combining PTP scores with any of the known cardiac risk factors (diabetes, hypertension, family history of CAD, tobacco use, chronic kidney disease) or coronary artery calcium score (CACS) score could further identify patients who might warrant additional cardiac testing at the Observation level of care (Sandoval et al., Circulation 2022, 146: 569-81; Allen et al., Circulation 2021, 143: 1659-72; Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

13:

Findings here refer to Pretest Probability (PTP) scores for patients with stable chest pain based on age, sex at birth, and symptoms (chest pain or dyspnea) for the clinical likelihood of a patient having obstructive coronary artery disease (CAD) (Knuuti et al., Eur Heart J 2020, 41: 407-77). Of note, dyspnea could be the main symptom or the only symptom. According to the guidelines published in 2019 by the European Society of Cardiology (ESC) and 2021 American multisociety guidelines, risk stratification derived from diagnostic testing and clinical assessment results in a strong recommendation for diagnosing CAD. PTP percentages are stratified into the following four risk categories:

- High: > 50%
- Intermediate: 16-50%
- Low: 6-15%
- Very low: ≤ 5%

Risk factors that can increase the likelihood of obstructive CAD include (Knuuti et al., Eur Heart J 2020, 41: 407-77; Reeh et al., Eur Heart J 2019, 40: 1426-35):

- Family history of CAD
- Tobacco use
- Chronic kidney disease
- Hypertension
- Diabetes
- Hyperlipidemia
- Increased body mass index
- Coronary artery calcium score (CACS) presence on CT scans

14:

Low risk defined as:

- HEART score ≤ 3 with negative serial troponins (high-sensitivity troponins (hs-cTn) at 0 and 1-3 hours or conventional troponins (cTn) at 0 and 3-6 hours)
- TIMI 0-1 (ADAPT study) with negative hs-cTn or cTn at 0h and 2 hours
- Negative (very low threshold) hs-cTn at 0 hours (on presentation) if symptom onset > 3 hours
- Negative (low threshold) hs-cTn at 0 and 2 hours

15:

Dyspnea may represent the predominant or only symptom suggesting acute coronary syndrome (ACS)

16:

Male sex assigned at birth

17:

Female sex assigned at birth

18:

This criterion refers to patients with the classification outlined in clinical decision pathways (CDP) as having intermediate-risk, patients without high-risk features, and those not classified as low-risk. These patients may have negative serial troponins and no evidence of myocardial infarction (MI) by electrocardiogram (ECG). In addition, some may have chronic but minimal or stable elevations of troponins. It is imperative to assess for prior testing in this group of patients. The patients in this risk category have been shown to have a probability of major adverse cardiovascular effects (MACE) of 16.6% at six weeks (Backus et al., Int J Cardiol 2013, 168: 2153-8). It may be appropriate for this group of patients to be evaluated in the hospital for additional testing in the Observation level of care, including a dedicated observation unit, if available. Intermediate risk score patients are defined as having any of the following (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285):

- HEART score 4-6
- TIMI score 2-4
- Not classified as low risk by a CDP and absence of high-risk features (e.g., elevated troponin consistent with acute myocardial injury, new ischemic ECG changes, new left ventricular dysfunction, new moderate-severe ischemia by stress testing, hemodynamic instability)

This criteria also includes patients with a known history of myocardial infarction and no obstructive coronary disease (MINOCA) or ischemia and no obstructive CAD (INOCA). This patient has the following characteristics:

- 60% (0.60) of cases are due to coronary microvascular dysfunction (CMD)
- Have elevated major adverse cardiovascular effects (MACE) including death, nonfatal MI, nonfatal stroke
- Have elevated risk of developing heart failure with preserved ejection fraction (HFpEF)
- Risk factors include: female sex assigned at birth, hypertension, diabetes, coronary flow reserve > 2 (CONFIRM trial)
- These patients can be managed medically if clinically appropriate (Herscovici et al., J Am Heart Assoc 2018, 7: e008868; Thygesen et al., J Am Coll Cardiol 2018, 72: 2231-64)

19:

Intermediate risk defined as:

- HEART score 4-6
- TIMI score 2-4
- Not classified as low risk by a clinical decision pathway (CDP) and absence of high-risk features (e.g., elevated troponin consistent with acute myocardial injury, new ischemic electrocardiogram changes, new left ventricular dysfunction, new moderate-severe ischemia by stress testing, hemodynamic instability)

20:

A non-diagnostic electrocardiogram may show nonspecific ST segment and T wave changes (ST segment deviation less than 0.5 millimeters and T wave inversion less than 2 millimeters). Also, a non-diagnostic electrocardiogram (ECG) should be compared to prior ECGs if available. Decision-making should not solely rely upon a normal or a non-diagnostic ECG. If a posterior wall infarct or a right ventricular infarct is suspected, additional leads (e.g., posterior with V7-V9, right-sided) should be checked for evidence of ischemic changes. Bundle branch blocks may mask underlying ischemia. In some patients with an initial normal ECG, serial ECGs should be repeated based on recurrent symptoms or other clinical change (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60; Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

21:

Instruction: Application of this criteria point is appropriate for patients with a left bundle-branch block (LBBB) which includes old or age indeterminant.

22:

Cardiac biomarkers are assessed in all patients with suspected acute coronary syndrome (ACS) and should include a cardiac troponin assay. Troponins are the gold standard for assessing myocardial injury; they can assist in diagnosing or excluding myocardial infarction and should be interpreted in conjunction with the patient's clinical symptoms, physical examination, and electrocardiogram findings. Historically, conventional troponin assays have had high specificities and sensitivities when used serially to identify myocardial infarction (MI). Over the last decade, however, there has been a shift towards using high-sensitivity troponins (hs-cTn). Numerous studies have shown a rapid, consistent, and significant improvement in diagnostic accuracy for detecting early MI at the time of emergency department (ED) presentation due to higher sensitivities and negative predictive values (NPV) of these newest generation troponin assays, especially when combined with algorithms such as clinical decision pathways (CDP) derived risk scores (Lipinski et al., Am Heart J 2015, 169: 6-16 e6; Januzzi et al., J Am Coll Cardiol 2019, 73: 1059-77). Acute MI is characterized by a dynamic blood level rise and (or fall, if the patient is presenting later) pattern above the 99th percentile of upper reference limit (URL) when combined with other characteristic presenting signs and symptoms (Thygesen et al., J Am Coll Cardiol 2018, 72: 2231-64). Troponin threshold levels defining a positive value may vary between laboratories and institutions (Allen et al., Circulation 2021, 143: 1659-72).

Based on the short time interval between the onset of chest pain and detectable blood levels of hs-cTn (due to its steep upslope), different rapid rule-in/rule-out protocols have been suggested and utilized. These protocols include (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60; Collet et al., Eur Heart J 2021, 42: 1289-367):

- 0h (if symptom onset was > 3 hours from ED presentation)
- 0h and 1-3 hour serial troponin
- HIGH-STEACS: initial hs-cTnI < 5 µg/L or hs-cTnT < 6 µg/L (if sympton onset was > 3 hours from ED presentation) or serial change from 0h to 3h is < 3 µg/L

Serial testing is encouraged to rule out MI, especially in early presenters (less than 3 hours) (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60). Overall, these algorithms have allowed for a statistically significant reduction in ED length of stay, an increased proportion of patients ruled out, and discharged from the ED compared to conventional troponins, which translates to substantial savings per year for US hospitals (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60; Anand et al., Circulation 2021, 143: 2214-24; Ibanez et al., Eur Heart J 2018, 39: 119-77; Thygesen et al., J Am Coll Cardiol 2018, 72: 2231-64; Twerenbold et al., J Am Coll Cardiol 2017, 70: 996-1012; Amsterdam et al., Circulation 2014, 130: e344-426).

23:

The 2019 European Society of Cardiology guidelines for chronic coronary syndromes recommend consideration of noninvasive cardiac testing in patients with stable chest pain with pretest probability (PTP) scores greater than 5% (0.05) for having obstructive coronary artery disease (CAD). Testing is most cost-effective when applied to patients with intermediate risk probability score for obstructive CAD. This refers to patients with a PTP of 16-50% (0.16-0.50) (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285; Knuuti et al., Eur Heart J 2020, 41: 407-77).

24:

Percutaneous coronary intervention (PCI) is the opening of a stenosed vessel by means of balloon angioplasty, stent insertion, atherectomy, or combination thereof (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

25:

Coronary computed tomography angiography (CCTA) is a reliable, noninvasive test with high negative predictive value used to rule out acute coronary syndrome (ACS) due to coronary artery disease (CAD). This testing is especially useful in patients with either intermediate risk based on clinical decision pathways or a pretest probability score of low to intermediate (Collet et al., Eur Heart J 2021, 42: 1289-367; Jiang et al., Cardiology 2014, 128: 304-12). Current guidelines support the use of CCTA in the evaluation of ACS, as an alternative or adjunct to stress testing (Gulati et al., Circulation 2021, 144: e368-e454). Advantages include high quality visualization of the coronary artery anatomy, the ability to exclude ACS due to CAD with a high degree of confidence, and brief study time. The use of CCTA may also reduce length of stay and hospital costs (Hamilton-Craig et al., Int J Cardiol 2014, 177: 867-73).

26:

This refers to the following cardiac testing modalities (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285):

- Exercise stress test
- Stress cardiac magnetic resonance imaging (MRI)
- Stress Echocardiogram (ECHO)
- Nuclear stress test (stress single photon emission computed tomography or SPECT) or cardiac stress-rest test
- Myocardial Perfusion PET Stress test

27:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

28:

Based on CDPs (clinical decision pathways), patients presenting to the emergency department (ED) with acute chest pain or anginal equivalent who are classified as high-risk should undergo cardiac catheterization. High-risk patients may also have moderate to severe ischemia seen on a stress test, new ischemic changes on electrocardiogram (ECG), new onset left ventricular dysfunction, or an abnormal troponin (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285). These groups of patients are at increased risk for short-term MACE of 50.1% at six weeks (Backus et al., Int J Cardiol 2013, 168: 2153-8).

29:

High risk defined as:

- HEART score 7-10
- TIMI score 5-7

30:

High risk patients can have the following criteria (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285):

- HEART score: 7-10
- TIMI: 5-7

31:

As per 2021 American multisociety guidelines, moderate to severe ischemia findings on stress imaging (e.g., stress single photon emission computed tomography (SPECT) or myocardial perfusion imaging) is defined as greater than 10% (0.10) ischemic myocardium placing these patients at higher risk for major adverse cardiovascular event (MACE) (e.g., death due to coronary artery disease, myocardial infarction) (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

32:

Defined as new ST depressions or T inversions on ECG

33:

- ST depressions
- T wave inversions

34:

This criteria point refers to a troponin-confirmed myocardial injury which may be a precursor to myocardial infarction. It includes patients with an elevated initial troponin (with no history of prior troponin elevations) and patients with a chronically elevated troponin with an acute troponin elevation above their usual baseline while awaiting repeat/serial testing.To establish a clinically significant elevation of serial troponins, a threshold change or delta is recommended depending on the type of troponin utilized to suggest myocardial injury with necrosis. Based on the literature, the following threshold changes or "deltas" are considered significant (Thygesen et al., Journal of the American College of Cardiology 2018, 72: 2231-64):

- Conventional troponins: > 20%(0.2) change
- High-sensitivity troponins:
  - If initial troponin ≤ 99th upper reference limit (URL), require > 50%(0.5) change
  - If initial troponin > 99th URL, require > 20%(0.2) change

35:

Therapeutic anticoagulation and dual antiplatelet therapy are recommended for patients with Non-ST-elevation myocardial infarction. Parenteral anticoagulation therapy includes unfractionated heparin, enoxaparin, bivalirudin, and fondaparinux. Bivalirudin is only recommended for invasive strategies. Fondaparinux should not be used as the only anticoagulant if a percutaneous coronary intervention is performed (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

36:

A non-ST segment elevation myocardial infarction (NSTEMI) is characterized by the clinical syndrome consisting of acute coronary syndrome (presentation with angina or anginal equivalents) by the presence of new electrocardiogram (ECG) ST-segment depression (e.g., suggestive of subendocardial injury pattern) or T wave inversion without ST elevation and positive troponins. Patients with NSTEMI are usually treated with dual anti-platelet therapy (DAPT), anticoagulation, and eventual cardiac catheterization if an invasive strategy is pursued. However, patients with troponin elevation due to demand ischemia or type 2 MI are usually managed by assessing and treating the underlying source of the increased demand (Collet et al., Eur Heart J 2021, 42: 1289-367) (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

37:

Instruction: Application of this criteria point is appropriate for patients whose chest, arm, jaw, or shoulder pain, or anginal equivalent, are either completely resolved or resolve following administration of medication.

38:

The guidelines standard of care for patients diagnosed with Non-ST elevation acute coronary syndrome (NSTE-ACS) consisting of unstable angina (UA) or non-ST elevation MI (NSTEMI) include the following unless contraindicated (Collet et al., Eur Heart J 2021, 42: 1289-367; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228):

- Beta blocker
- Aspirin
- Antiplatelet
- Anticoagulant
- Angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blocker (ARB) if heart failure present
- Statin (high intensity if clinically appropriate)
- Supplemental oxygen for O2 saturation below 90% (0.90)
- Nitrates for ongoing chest pain

Patients admitted with NSTE-ACS are appropriate for treatment at the Intermediate level of care for immediate stabilization of symptoms, relief of ischemia with standard medical therapy, bed rest, and further testing with serial troponins and electrocardiograms. Depending on symptoms and findings, patients are managed with either an invasive or selective invasive strategy. The average length of stay has decreased in recent years due to earlier interventions (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

Invasive Strategy:

This criterion refers to those patients diagnosed with NSTEMI who will require cardiac catheterization during the current admission. According to 2020 European Society of Cardiology (ESC) guidelines for NSTEMI and multiple randomized controlled trials and meta-analyses, a strong recommendation was given to this group to undergo diagnostic cardiac catheterization within 24 hours based on clinical findings as outlined below. This approach is referred to as early invasive strategy. In the 2014 American College of Cardiology (ACC) guidelines, invasive strategy was also described; however, recommendations allowed for the timing of invasive cardiac intervention from 2 hours for up to 72 hours (Amsterdam et al., Circulation 2014, 130: e344-426). This strategy also allows for the stabilization of plaque with the use of parental anticoagulants. Up to 72 hours was referred to as a delayed invasive strategy compared to immediate (less than 2 hours) and early invasive strategies (2-24 hours). Nevertheless, all of the strategies outlined by the ACC and ESC guidelines, particularly the early invasive strategy, facilitate early risk stratification, earlier revascularization, and earlier discharge. The timing of cardiac intervention depends on multiple clinical factors (Collet et al., Eur Heart J 2021, 42: 1289-367; Kofoed et al., Circulation 2018, 138: 2741-50; Milasinovic et al., Atherosclerosis 2015, 241: 48-54; Mehta et al., N Engl J Med 2009, 360: 2165-75):

- Immediate < 2h: heart failure related to NSTEMI, new mitral regurgitation after myocardial infarction, hemodynamic instability, sustained VT/VF, refractory or recurrent angina or ischemia at rest. (strong recommendation)
- Early ≥ 2h to ≤ 24h: dynamic or new ST-segment changes, resuscitated cardiac arrest without cardiogenic shock or ST elevations (strong recommendation), established diagnosis of NSTEMI

Selective Invasive Strategy:
Some patients with NSTEMI may be medically managed with the addition of antiplatelet agents (e.g., clopidogrel, prasugrel, ticagrelor) and/or aspirin. According to the 2020 ESC guidelines, a strong recommendation was given to this category of patients designated as selective invasive strategy (formerly referred to as conservative strategy and ischemia-guided strategy). These patients include those whose conditions have stabilized with medical therapy and may not need invasive coronary angiography. However, according to the 2014 ACC guidelines, a strong recommendation was given to complete a non-invasive evaluation for severe ischemia occurring at low thresholds of stress before discharge after NSTEMI (Amsterdam et al., Circulation 2014, 130: e344-426). This strategy may be appropriate for patients with low-risk scores (TIMI 0-1), negative troponins, female sex assigned at birth, and the absence of high-risk features. If symptoms do recur or non-invasive testing reveals ischemia, cardiac catheterization may be warranted. This strategy may be selected for certain patients diagnosed with NSTEMI who could have an increased risk of developing complications from invasive testing such as cardiac catheterization (e.g., older patients, chronic kidney disease, allergy) if clinically warranted (Collet et al., Eur Heart J 2021, 42: 1289-367).

39:
Unstable angina (UA) is defined by clinical (e.g., chest discomfort or anginal equivalent) and electrocardiographic (e.g., ST depression or prominent T wave inversion) evidence of acute myocardial ischemia in the absence of positive troponins. UA presents as (Knuuti et al., Eur Heart J 2020, 41: 407-77):

- Rest angina: characteristic pain occurring at rest or for prolonged period (typically lasting over 20 minutes)
- New onset angina: recent (< 2 months) onset of moderate to severe angina
- Crescendo angina: progressively increases in frequency and intensity, at lower thresholds, over brief period of time.

40:
The American College of Cardiology Foundation/American Heart Association (ACCF/AHA) defines angina utilizing the Canadian Cardiovascular Society (CCS) classification scale (Owlia et al., J Am Heart Assoc 2019, 8: e012811; Campeau, Circulation 1976, 54: 522-3):

- Class I: Absence of angina with ordinary physical activity (e.g., occurs only in the setting of significantly increased demands such as strenuous exercise)
- Class II: Slight limitation of ordinary physical activity (e.g., angina occurs upon walking more than 2 blocks or climbing more than 1 flight of stairs at a normal pace)
- Class III: Marked limitation of ordinary physical activity (e.g., angina occurs upon walking 1 to 2 blocks or climbing 1 flight of stairs at a normal pace)
- Class IV: Angina occurs with any physical activity (Knuuti et al., Eur Heart J 2020, 41: 407-77)

41:
Instruction: Patients with persistent acute chest pain, anginal equivalent, or those with silent ischemia should be managed at the Critical level of care (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285; Ibanez et al., Eur Heart J 2018, 39: 119-77; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

42:
The diagnostic value of nonspecific ST-segment changes of less than 0.5 millimeters and T-wave inversion less than 2 millimeters is low. A strong suspicion of acute myocardial ischemia is based on the following thresholds:

- ST-segment depression of 0.5 millimeters or greater

- T-wave inversion of 2 millimeters or greater

43:

Instruction: Application of this criteria point is appropriate for new electrocardiogram (ECG) changes, changes from a baseline ECG, or when the baseline is unknown.

44:

In patients with pacemakers, it may be difficult to assess for critical ST-segment changes due to the pattern resulting from the paced rhythm. The modified Sgarbossa criteria can be utilized to assist in the diagnosis of myocardial infarction in this circumstance (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60).

45:

Instruction: In the rare instance when there is an absolute contraindication to anticoagulation for a patient with acute coronary syndrome (i.e., recent surgery, active GI bleed, very low platelets), secondary review would be appropriate.

46:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

47:

The presence of a left bundle branch block (LBBB), a coronary conduction abnormality, can make it challenging to diagnose ST- segment elevation myocardial infarction (STEMI) as it can mask the typical electrocardiographic evidence. In suspected acute coronary syndrome, any new, or presumed new LBBB could be considered STEMI equivalent or pathological, and warrants immediate evaluation (Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60).

48:

This criterion pertains to patients who develop new-onset arrhythmia after a myocardial infarction (MI). Not all post-MI arrhythmias require intervention. The incidence of malignant ventricular arrhythmias (VA) depends on the infarct size. Treatment for life-threatening ventricular fibrillation or sustained ventricular tachycardia includes immediate defibrillation or cardioversion. Temporary pacing may be indicated for certain types of bradyarrhythmias, such as atrioventricular block, complete heart block, or new bundle-branch block, especially when the arrhythmia causes hemodynamic instability. Initiation of antiarrhythmics may be helpful due to ongoing coronary ischemia. For the occurrence of atrial fibrillation in the post-MI setting, urgent cardioversion may be recommended, especially in the presence of hemodynamic instability (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).

49:

Intravenous nitroglycerin is indicated for persistent angina, heart failure, or hypertension (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).
Instruction: Use of this criteria point is intended for continual intravenous nitroglycerin, and excludes sublingual, paste, or nitroglycerin patch.

50:

An ST-elevation myocardial infarction (STEMI) is characterized by the presence of new electrocardiogram (ECG) changes with ST-segment elevations and positive troponins (Ibanez et al., European heart journal 2017, 39: 119-77).

51:

The guideline standards of care for patients with ST-segment elevation myocardial infarction (STEMI) recommend the following unless contraindicated (Ibanez et al., European heart journal 2017, 39: 119-77; O'Gara et al., Circulation 2013, 127: e362-425):

- Angiotensin-converting enzyme (ACE) inhibitors for patients with heart failure, left ventricular ejection fraction (LVEF) of 40%(0.40) or less, diabetes mellitus, or an anterior infarct.
- Angiotensin receptor blocker (ARB) for patients with heart failure, left ventricular ejection fraction (LVEF) of 40%(0.40) or less, who are intolerant of ACE inhibitors.
- Aldosterone blockers for patients with heart failure, left ventricular ejection fraction (LVEF) of 40%(0.40) or less, or diabetes mellitus, who are already taking an ACE inhibitor or beta-blocker if hyperkalemia or renal failure are not present.
- Beta blockers administered promptly in the setting of confirmed acute coronary syndrome (ACS) can reduce heart rate, thus reducing myocardial oxygen demand, systemic arterial pressure, and myocardial contractility. Beta blockers can reduce infarct size, the risk of heart failure, ventricular arrhythmias, and cardiogenic shock. However, there are mixed data regarding the impact of early beta blocker use on mortality (Nicolau et al., Cardiovascular drugs and therapy 2018, 32: 435-42; Park et al., Am J Med 2014, 127: 503-11; Pizarro et al., J Am Coll Cardiol 2014, 63: 2356-62; Chatterjee et al., Int J Cardiol 2013, 168: 915-21; Hirschl et al., Crit Care Med 2013, 41: 1396-404). Beta blockers may be administered intravenously in the setting of tachyarrhythmia, hypertension, or ongoing ischemia. In patients who are not appropriate for beta blocker therapy and do not have evidence of significant left ventricular (LV) dysfunction, a nondihydropyridine calcium channel blocker should be initiated (Ibanez et al., European heart journal 2017, 39: 119-77; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).
- Nondihydropyridine calcium channel blockers may be used in the treatment of ongoing or recurrent ischemic symptoms in patients:
    - Already receiving adequate doses of nitrate and beta blockers
    - Unable to tolerate adequate doses of nitrates or beta blockers
    - With variant angina
    - With hypertension and recurrent unstable angina

- Aspirin should be administered as soon as possible and continued indefinitely unless contraindicated. Clopidogrel or prasugrel can be used for those patients who are unable to tolerate aspirin due to hypersensitivity or major gastrointestinal intolerance. Studies have shown long-term prevention with lower doses (75-100 mg) have been similarly effective as higher doses with less adverse events (Ibanez et al., European heart journal 2017, 39: 119-77; Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228).
- Antiplatelet therapy includes administration of a $P2Y_{12}$ receptor inhibitor (e.g., clopidogrel, prasugrel, ticagrelor), or in some instances, a glycoprotein IIb/IIIa inhibitor (e.g., eptifibatide, tirofiban). Prasugrel has been shown to be more effective than clopidogrel but is associated with a significantly higher risk of bleeding. It is contraindicated in patients with a history of transient ischemic attack (TIA) or stroke and is typically not recommended in patients over 75 years of age. Patients weighing less than 60 kilograms have a further increased risk of bleeding on prasugrel. Registry data suggest that prasugrel may be associated with a lower mortality risk in patients with cardiogenic shock. Ticagrelor has also been shown to be more effective than clopidogrel, but at the expense of increased intracranial hemorrhage and stroke (Udell et al., JACC Cardiovasc Interv 2014, 7: 604-12; Orban et al., Thromb Haemost 2014, 112: 1190-7; O'Gara et al., Circulation 2013, 127: e362-425; Melloni et al., 2013 In: Antiplatelet and Anticoagulant Treatments for Unstable Angina/Non-ST Elevation Myocardial Infarction). The administration of both a $P2Y_{12}$ receptor inhibitor and a glycoprotein IIb/IIIa inhibitor should be reserved for high-risk patients in whom the benefits outweigh the risk of bleeding, specific rescue situations with presence of higher intraprocedural thrombus burden, or slow-flow closure in stented coronary arteries (Neumann et al., Eur Heart J 2019, 40: 87-165; Ibanez et al., European heart journal 2017, 39: 119-77; Amsterdam et al., Circulation 2014, 130: e344-426).

52:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported. Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

53:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

54:

An intra-aortic balloon pump (IABP) is a mechanical circulatory support device that utilizes a catheter-mounted intravascular device positioned in the descending thoracic aorta. The balloon automatically inflates during early diastole, increasing coronary blood flow and deflates during early systole, reducing afterload as the left ventricle ejects. Indications for use include cardiogenic shock, temporary support of left ventricular function after myocardial infarction (MI), preservation of myocardial viability, or refractory ventricular arrhythmias post-MI or preoperative bridge before coronary artery bypass graft (CABG) surgery. Contraindications include aortic regurgitation and aortic dissection.

55:

Placement of an intra-aortic balloon pump (IABP) may be beneficial for patients with recurrent ischemia despite intensive medical management, mechanical complications of a myocardial infarction (MI), hemodynamic instability, or cardiogenic shock awaiting coronary angiography and/or revascularization (Amsterdam et al., Circulation 2014, 130: 2354-94; Anderson et al., J Am Coll Cardiol 2013, 61: e179-347; O'Gara et al., Circulation 2013, 127: e362-425; Thiele et al., N Engl J Med 2012, 367: 1287-96).

56:

Thrombolytic agents are medications that act to dissolve culprit clots. Indications for the use of thrombolytic agents include acute myocardial infarction, thrombotic stroke, massive pulmonary embolism, deep vein thrombosis, peripheral arterial thromboembolic occlusion, and vascular graft thrombosis. The most common complication of thrombolytic therapy is hemorrhage. Risk increases with the concomitant use of heparin or other antithrombotic agents. There is also a risk of embolism occurring from a dissolving clot.

57:

A percutaneous left ventricular assist device (VAD) is used to protect the heart during invasive procedures or temporarily for short-term mechanical circulatory support while treatment decisions are made.

58:

In some facilities a ventricular assist device (VAD) will be used as a bridge to heart or heart-lung transplant. This may stabilize a patient long enough to survive until transplant, but the patient is still fragile and relatively unstable. **Instruction**: In this clinical setting, the patient may require Critical care until a donor organ is available. Due to the restrictions associated with these devices, the Critical level of care is required for percutaneous VADs.

59:

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

60:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

61:

When patients with suspected acute coronary syndrome (ACS) or ischemic heart disease have a nonischemic or unchanged electrocardiogram (ECG) and negative troponins, a stress test (e.g., pharmacological, exercise) or cardiac imaging was previously recommended to be performed within 72 hours of discharge (Amsterdam et al., J Am Coll Cardiol 2014, 64: e139-228). According to the 2021 American multisociety guidelines, low-risk patients by clinical decision pathways (CDP) or serial troponins who are clinically stable and have less than 1% frequency of major adverse cardiovascular effect (MACE) within 30 days may be considered for discharge with close outpatient follow up (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

62:

An objective cardiac risk assessment may include an exercise or pharmacologic stress test (with or without nuclear imaging), coronary computed tomography angiography (CCTA), stress echocardiogram, or cardiac catheterization. The choice of modality may be influenced by the patient's symptoms, functional status, local expertise, and availability (Gulati et al., Circulation 2021, 144: e368-e454).

63:

This criteria point refers to the patients who had negative troponins and underwent cardiac diagnostic testing before their discharge at the Observation level of care. These patients were categorized as low or intermediate-risk. The 2021 American multisociety guidelines recommend discharge for those patients when they also meet one of the parameters below (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285):

1. Patients with no prior outpatient cardiac testing, intermediate-risk by clinical decision pathway (CDP), no known coronary artery disease (CAD), and:
   - Stress test with negative or mildly abnormal findings
   - Coronary computed tomography angiography (CCTA) showing < 50% (0.50) obstruction
   - CCTA showing ≥ 50% (0.50) obstruction with a shared decision to treat medically

2. Patient with prior cardiac testing and:
   - Stress or CCTA completed in the warranty period and no change to the pattern, type, or duration of symptoms
   - CCTA showing < 50% (0.50) obstruction after known mildly abnormal or inconclusive stress test

3. Patients with prior history of nonobstructive CAD and:
   - No change on CCTA
   - CCTA now with ≥ 50% (0.50) obstruction and no moderate or severe abnormality on stress test

4. Patients with obstructive CAD and normal stress test results

64:

This criteria point refers to patients who are categorized as low-risk and have at least a low (6-15%) pretest probability of obstructive coronary artery disease (CAD) and were admitted at the Observation level of care. This criteria point also refers to patients with higher clinical suspicion requiring a third set of troponins to better evaluate the trend in their rise. If the patient and the clinician(s) have reached a shared decision for timely outpatient cardiac testing, it is recommended to complete the necessary cardiac testing within 30 days (Gulati et al., Circulation 2021, 144: e368-e454).

65:

This criteria refers to abnormal findings or results on cardiac diagnostic testing that warrants further invasive testing (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285):

- Stress testing with moderate to severe abnormalities
- Functional flow reserve CT (FFR-CT) ≤ 0.8
- Coronary computed tomography angiography (CCTA) with findings consistent with high-risk coronary

artery disease (CAD): left main disease ≥ 50% (0.50), proximal left anterior descending artery (LAD) disease or multivessel disease (≥ 70% (0.70) stenosis)

66:

Admission to the Intermediate level of care is indicated in patients with acute coronary syndrome (ACS) confirmed by a positive stress test or coronary computed tomography angiography (CCTA) (Amsterdam et al., Circulation 2014, 130: 2354-94).

67:

- Stress testing with moderate to severe abnormalities
- Functional flow reserve CT (FFR-CT) ≤ 0.8
- Coronary computed tomography angiography (CCTA) with findings consistent with high-risk coronary artery disease (CAD): left main disease ≥ 50% (0.50), proximal left anterior descending artery (LAD) disease or multivessel disease (≥ 70% (0.70) stenosis)

68:

This criteria line refers to those patients who present with acute chest pain or anginal equivalent and have high risk findings such as new ischemic changes on electrocardiogram (ECG), new moderate to severe ischemia on stress test, new echocardiogram findings of an ejection fraction (EF) ≤ 40% (0.40), or a high risk score (e.g., HEART, TIMI score) based on clinical decision pathways. For these patients, guidelines recommend cardiac catheterization (Gulati et al., J Am Coll Cardiol 2021, 78: e187-e285).

69:

Instructional: This refers to patients that were admitted with ST-elevation myocardial infarction (STEMI) on episode day 1.

70:

Percutaneous coronary intervention (PCI) is the opening of a stenosed coronary vessel by means of balloon angioplasty, stent insertion, atherectomy, catheter thrombectomy, or a combination thereof.

71:

Instruction: This criteria point is appropriate for patients having undergone a catheterization, whether or not a percutaneous coronary intervention (PCI) was performed.

72:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

73:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

74:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

75:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

76:

Hypertension is defined as a systolic blood pressure (SBP) of greater than 130 mmHg or a diastolic blood pressure (DBP) of 80 mmHg or greater based on 2017 Hypertension Clinical Practice Guidelines. Recommendations for initial management in a post-myocardial infarction (MI) period would include dietary salt restriction, anti-hypertensive agents (e.g., beta blockers, Angiotensin-converting enzyme (ACE) inhibitors, Angiotensin II receptor blockers (ARBs), calcium channel blockers (CCB)) where clinically appropriate and education on lifestyle modifications. The goal of antihypertensive therapy is to reduce morbidity and mortality. Goals for target BP level should be individualized, but a SBP less than 130 mmHg and a DBP less than 80 mmHg is desirable (Casey et al., Circ Cardiovasc Qual Outcomes 2019, 12: e000057; Kontos et al., J Am Coll Cardiol 2022, 80: 1925-60; Whelton et al., Hypertension 2018, 71: 1269-324).

77:

Relevant complications or comorbidities are conditions that require active management during an inpatient stay.

78:

The anticoagulation regimen may include low molecular weight heparin (LMWH), unfractionated heparin (UFH), or fondaparinux with or without warfarin, or warfarin alone. For adult patients, non-vitamin K oral anticoagulants (NOACs) (also known as novel oral anticoagulant, direct oral anticoagulant (DOAC), and target-specific oral anticoagulant) such as dabigatran, rivaroxaban, apixaban, or edoxaban, may also be used (Ortel et al., Blood Adv 2020, 4: 4693-738).

79:

Warfarin dosage is adjusted according to the following international normalized ratio (INR) values (Kernan et al., Stroke 2014, 45: 2160-236; Nishimura et al., Circulation 2014, 129: e521-643; Whitlock et al., Chest 2012, 141: e576S-600S):

- Range of 2.0-3.0 with a target of 2.5 for patients with bioprosthetic valves, mechanical aortic valves, or for patients without implanted mechanical valves
- Range of 2.0-3.0 with a target of 2.5 for patients with ischemic stroke or transient ischemic attack (TIA) with a history of rheumatic mitral valve disease and atrial fibrillation
- Range of 2.5-3.5 with a target of 3.0 for patients with mechanical mitral valves or combination mitral/aortic valve implantation

80:

Instruction: For these criteria, medication adjustment refers to a change in dose, frequency of administration, or change in medication.

81:

Initial refers to the first time a medication or treatment is utilized. During the period of initiation, if the medication or treatment is temporarily discontinued up to 24 hours (therapeutic pause) it is still considered initial. If a tolerated medication or treatment is discontinued for more than 24 hours and then restarted, it is not considered initial. Medications that were initiated on an outpatient basis and are discontinued prior to hospitalization may be considered initial based on the need for monitoring or duration of discontinuation. Medications that were temporarily held due to concurrent illness and are resumed are not considered initial.

82:

Clinical expert validation of best practice shows that in the setting of acute coronary syndrome (ACS) amiodarone and sotalol are the safest and most widely used oral antiarrhythmic medications. Other oral antiarrhythmic medications may prolong the QT interval or have the potential to cause significant harm in the setting of ACS. Drugs with a high proarrhythmic risk may cause worsening of an existing arrhythmia or precipitate a new one. Cardiac monitoring is recommended during initiation of these medications.

83:

Instruction: Examples of antiarrhythmics with proarrhythmic risk include, but are not limited to, amiodarone, disopyramide, dofetilide, dronedarone, flecainide, mexiletine, propafenone, and sotalol. Many factors influence the appropriate choice of drug, including underlying medical conditions and concurrent drug administration. In some instances, such as patients who are pregnant or are breastfeeding, there are drugs which are contraindicated or have limited evidence to support their use.

84:

This criterion refers to the patient diagnosed with a new intracardiac thrombus after an ST-elevation myocardial infarction (STEMI). The incidence of a left ventricular (LV) thrombus after an anterior wall STEMI ranges between 4-39%. Larger infarcts pose higher threats to the development of left ventricular thrombus. Due to improvements in reperfusion and anti-thrombotic interventions, there has been a steady decline; however, the risk of development remains significant. Studies have shown a higher risk of cerebral or arterial embolization and major adverse cardiovascular events (MACE) (including all-cause death, and myocardial infarction (MI)) at 22% and 37%, respectively, while subsequently affecting mortality. Anticoagulation therapy continues to be the mainstay treatment after diagnosing intracardiac thrombus to reduce the complications described earlier. Upon diagnosis, current guidelines continue to recommend initiation of heparin infusion or low molecular weight heparin (LMWH) bridge with the initiation of vitamin K antagonist (VKA) such as warfarin for a total duration of 3-6 months. At the same time, patients should be continually assessed for potential bleeding risk when there is concomitant use of antiplatelet agents after an MI. The 2021 American Heart Association/American Stroke Association (AHA/ASA) guidelines highlighted patients with a history of stroke or transient ischemic attack (TIA) and new LV thrombus for therapeutic warfarin for greater than 3 months to reduce the incidence of stroke. There is limited evidence for using direct oral anticoagulants (DOACs) for treating intracardiac thrombus based on small randomized controlled trials. Studies have shown conflicting evidence regarding the efficacy and safety of DOACs vs. VKA. In situations where VKAs cannot be tolerated or time to therapeutic INR (TTR) continually remains less than 50% (0.50), DOACs should be highly considered. The incidence of embolization due to poorly controlled anticoagulation was 19%. 2022 AHA statement outlines DOACs as a reasonable alternative to VKA, especially in whom therapeutic INR range may be challenging to achieve or the presence of social determinants of health (SDOH) and associated barriers (e.g., outpatient services unavailable). Use of dabigatran or edoxaban requires at least five days of parental anticoagulation before oral therapy alone (Camaj et al., J Am Coll Cardiol 2022, 79: 1010-22) (Levine et al., Circulation 2022, 146: e205-e23) (Kleindorfer et al., Stroke 2021, 52: e364-e467; McCarthy et al., JAMA Cardiol 2018, 3: 642-9).

This criterion also refers to the patient requiring anticoagulation for new onset intracardiac thrombus with a history of or further suspicion of heparin-induced thrombocytopenia (HIT). HIT results from sensitization to the heparin molecule and usually occurs between 5 to 10 days after the current heparin exposure begins but may occur as early as minutes to hours after the start of heparin therapy. HIT's hallmark is a greater than 50% (0.50) serial reduction of the patient's baseline platelet count. Parenteral direct thrombin inhibitors (DTI) (strong recommendation) are recommended in patients with active HIT, a history of HIT, or those with a strong clinical suspicion of HIT who require anticoagulation. Low doses of warfarin may be restarted during DTI therapy once the platelet count has significantly improved (usually platelet count greater than 150,000) and there is clinical evidence of improvement in the patient's thrombosis. DTI therapy should be continued until the platelet count is stabilized. A 5-day overlap with warfarin and DTI therapy is recommended (Cuker et al., Blood Adv 2018, 2: 3360-92).

85:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO_2) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO_2 with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO_2 retention, where a precise inspired FiO_2 is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO_2. The following are estimates of O_2 delivered at the associated flow rates:

LOW-FLOW SYSTEMS

Nasal cannula

Room air 21%(*0.21*)
1L 24%(*0.24*) **4L** 36%(*0.36*)
2L 28%(*0.28*) **5L** 40%(*0.40*)
3L 32%(*0.32*) **6L** 44%(*0.44*)

Simple oxygen masks

- 35-50%(*0.35-0.50*) FiO_2 (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO_2 in the mask

Venturi masks

- 24-50%(*0.24-0.50*) FiO_2 (4-12 L/min)

Partial rebreathing masks

- 40-70%(*0.40-0.70*) FiO_2 (6-10 L/min)

Non-rebreathing masks

- 60-80%(*0.60-0.80*) FiO_2 (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS

Air-entrainment masks/nebulizers

- 24-40%(0.24-0.40) FiO_2

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO_2

86:

This criterion refers to the patient who was found to have ventricular arrhythmia (VA; ventricular fibrillation or ventricular tachycardia) or out-of-hospital cardiac arrest before, during, or after a myocardial infarction (MI). Most of life-threatening VAs are related to ischemic heart disease, particularly in older patients. According to 2015 European Society of Cardiology (ESC) and 2017 American College of Cardiology/American Heart Association (AHA/ACC) guidelines, the use of implantable cardioverter-defibrillator (ICD) has shown an improvement in survival post-MI with certain conditions to prevent sudden cardiac death (Al-Khatib et al., J Am Coll Cardiol 2018, 72: e91-e220)(Priori et al., Eur Heart J 2015, 36: 2793-867). These include LVEF ≤ 35% (0.35) with ischemic heart disease or cardiomyopathy; other risk factors for peri-ACS ventricular arrhythmias include chronic obstructive pulmonary disease (COPD), prior MI with subsequent non-sustained ventricular tachycardia (NSVT) and hypertension. The 2017 ACC guidelines give a strong recommendation for implantation of an ICD in those post-MI patients with left ventricular ejection fraction of ≤ 35% (0.35) and class II or III heart failure and more than 40 days after the MI (Al-Khatib et al., J Am Coll Cardiol 2018, 72: e91-e220).

87:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

88:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

89:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

90:

For most patients with acute coronary syndrome, medications used to control ischemia during the hospitalization should be continued following discharge. The medication regimen should be individualized based on diagnostic findings, risk factors, medication tolerance, and procedural interventions (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414).

91:

Prior to discharge from the hospital, patients with acute coronary syndrome should be informed about symptoms of worsening ischemia, and should be instructed when and where they should seek emergency care when such symptoms occur. Patients should be instructed that anginal discomfort lasting longer than two or three minutes requires action. The patient should discontinue physical activity, or remove themselves from the stressful situation. Pain that does not subside immediately should be treated with sublingual nitroglycerin up to three doses or as prescribed. The EMS system should be activated for pain that is not relieved after the first dose of sublingual nitroglycerin (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414; Anderson et al., J Am Coll Cardiol 2013, 61: e179-347).

92:

Secondary risk prevention, including aggressive risk factor management through medical therapy and lifestyle modification, is the mainstay of therapy for patients recovering from acute coronary syndrome. Lifestyle and risk factor modification includes stress reduction, lipid management, optimizing diet and exercise, weight management, and smoking cessation if indicated. Cardiac rehabilitation programs are recommended to improve long-term outcomes, including functional capacity, morbidity and mortality rates, symptom control, and quality of life. Cardiac rehabilitation involves assessment, surveillance, counseling, and training to ensure lifestyle and risk factor modifications are maintained (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414).

93:

A high-intensity statin should be administered once the patient is stabilized, provided no contraindications are present. Statins reduce the risk of death due to coronary heart disease, recurrent myocardial infarction, revascularization needs, and stroke. Contraindications to statin therapy include hypersensitivity and active liver disease (Benjo et al., Catheter Cardiovasc Interv 2015, 85: 53-60; Amsterdam et al., Circulation 2014, 130: 2354-94; O'Gara et al., Circulation 2013, 127: 529-55).

94:

Physical activity can reduce symptoms in cardiovascular disease as well as improve other cardiac risk factors by aiding weight loss and weight management. Patients should be encouraged to take daily walks immediately following discharge. In patients with uncomplicated acute coronary syndrome, sexual activity can be resumed within 7 to 10 days and driving can begin 7 days after discharge. In patients who have had a complicated recovery, these determinations should be made on a case-by-case basis. Enrollment in a cardiac rehabilitation program after discharge may enhance blood pressure control, exercise capacity, patient education, compliance and encourage participation in a regular exercise program (Arnett et al., J Am Coll Cardiol 2019, 74: 1376-414). The data are mixed as to whether center-based cardiac rehabilitation increases physical activity; home-based cardiac rehabilitation may lead to increased physical activity (Ter Hoeve et al., Phys Ther 2015, 95: 167-79). Patients who attend early, short-term, comprehensive cardiac rehabilitation programs may have improved prognoses (Rauch et al., Eur J Prev Cardiol 2014, 21: 1060-9).

95:

The occurrence of depression and anxiety are common following myocardial infarction and can be associated with an increase in behaviors that further increase cardiovascular risk (e.g., smoking, alcohol consumption, over-eating, medication non-adherence) (Berg et al., Journal of psychosomatic research 2018, 112: 66-72). Upon discharge patients should be made aware of the warning signs of depression and anxiety and the importance of seeking out help from a mental health professional should symptoms arise.

96:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

97:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

98:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

99:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

100:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

101:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.
Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

102:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

103:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)

- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

104:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

105:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

106:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

107:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

108:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations

- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

109:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

110:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.

111:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

112:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).